

# Cade Clonts

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## SUMMARY

Electrical engineering graduate (Magna Cum Laude) in ASU's accelerated M.S. program, focused on embedded hardware and microelectronics. Experience in PCB design, embedded firmware, and signal processing, with DoD-supported SCALE research and electronic packaging coursework. U.S. Citizen; eligible for DoD security clearance; open to relocation.

## EDUCATION

**M.S., Engineering (Accelerated Master's)** — Arizona State University, Mesa, AZ **Expected Spring 2027**

*Graduate coursework:* Power Electronic Converters & Systems, Engineering Analysis I, Multimodal LLMs for Engineering Applications

**B.S.E., Engineering (Electrical Systems)** — Arizona State University, Mesa, AZ **May 2026**

**GPA 3.62 • Magna Cum Laude • Dean's List (4 terms)**

*Relevant coursework:* Heterogeneous Integration & Electronic Packaging, Analog-Digital Interface, Principles of Modern Electromagnetism, Embedded Systems Design I & II, Principles of Systems Engineering (graduate)

## TECHNICAL SKILLS

**Hardware:** Altium • KiCad • PCB Bring-up • SMT Soldering • Buck/LDO Rail Design • Oscilloscope • DMM

**Embedded & Software:** C • Python • MicroPython • MATLAB • ROS • ESP32 • PSoc • UART • I<sup>2</sup>C • PWM • ADC • MQTT/TLS

**Microelectronics & Fab:** SEM • Chip Cross-Sectioning & Polishing • MSLA/FDM 3D Printing • AutoCAD • Welding

## PROFESSIONAL EXPERIENCE

**Student Researcher, ASU 3DX Research Group (SCALE)** **Nov 2023 – Present**

- Lead a study of 3D-printed branching capillary channels for electronics cooling, testing Murray's, da Vinci's, and Hack's laws
- Selected the group's high-resolution MSLA printer (Elegoo Saturn 4 Ultra 16K) within a \$1,000 budget for hair-like structures
- Compiled the additive manufacturing process comparison for a co-authored SFF 2026 conference paper

**East Valley Industrial Products (part-time)** **Aug 2020 – Present**

- Fabricate diamond-plate steel water box lids: plasma-cut ~12 blanks per 4×8 ft sheet, bend lips, weld ribs
- Cross-trained across the full production process; currently responsible for finishing and painting

**Case IH, Goodman AG** **Nov 2020 – Jan 2023**

- Installed, calibrated, and repaired GPS autopilot systems on farm tractors
- Diagnosed and resolved autopilot faults on-site, minimizing downtime for growers

## ACADEMIC PROJECTS

**ESP32 Wi-Fi/MQTT Gateway PCB** **EGR 314 • Spring 2025**

- Designed a custom ESP32-S3 board in Altium as the Wi-Fi gateway for a networked multi-board system, using trade studies to select the MCU and regulator
- Defined a 64-byte framed UART protocol with addressing, broadcast, and forwarding; wrote async MicroPython firmware publishing sensor data over MQTT with TLS mutual authentication
- Designed dual power inputs (12 V synchronous buck, 5 V USB LDO) from a load budget with 25% margin; demoed end to end at the Fulton Schools Innovation Showcase

**Digital Filter Design** **EGR 334 • Fall 2025**

- Designed a Butterworth IIR digital filter in MATLAB and analyzed its magnitude and phase response
- Compared it against the team's Chebyshev design, weighing Butterworth's flat passband against Chebyshev's steeper roll-off

**ROS Computer Vision Color-Tracking Robot** **EGR 530 • Spring 2026**

- Built a computer vision application in ROS on a ROSMASTER X3 mobile robot that detects a target by color in the onboard camera feed and drives the robot to follow it

## PUBLICATIONS

I. Chavez Martinez, C. Yuen, **C. Clonts**, Z. Okun, A. Sarrasin, A. Potts, M. Nunez, A. Nizamudeen, R. Duong, H. Emady, C. Ozturk, D. Bhate, "Additive Manufacturing of Hair-like Materials: Design Principles, Process Constraints, and Manufacturing Strategies," accepted for publication in *Proceedings of the 37th Annual International Solid Freeform Fabrication Symposium*, Austin, TX, Aug. 3–5, 2026.